

Owadokun Tosin Tobi

Senior AI Engineer | MLOps & LLM Evaluation Specialist Lagos, Nigeria | 08077228021 |
tosinowadokun11@gmail.com GitHub: github.com/eatosin | LinkedIn: linkedin.com/in/tosinowadokun

Professional Summary

Physics undergraduate and AI Engineer specializing in LLM Evaluation, MLOps, and Deep Learning. Leveraging experimental physics principles to build rigorous, self-healing AI infrastructure. Engineered 7+ production-grade systems including adversarial evaluation pipelines, autonomous agents, and real-time recommendation engines. Passionate about bridging the gap between scientific research and scalable, fault-tolerant software engineering.

Technical Skills

- **Core:** Python (Pandas, NumPy, Scikit-Learn), PyTorch, XGBoost, Git/GitHub.
- **AI Engineering:** LLM Evaluation (RLHF), RAG Pipelines, Agentic Workflows (LangGraph), Recommender Systems (SASRec), Audio Forensics (CNNs).
- **MLOps & Cloud:** FastAPI, Docker, Microservices, Supabase, Render/Hugging Face, CI/CD.
- **Tools:** Google Gemini 2.5, Tavily API, Streamlit, Pydroid 3 (Mobile-Edge Development).

Featured Projects

NewsAgent Pro | Autonomous Multi-Modal Content Engine [Live Demo]

- Architected an agentic workflow using LangGraph and Tavily API to autonomously research, draft, and publish viral news content.
- Integrated Flux.1 (Image Gen) and Python Pillow to programmatically generate and brand "Newsflash" visual assets in real-time.
- Implemented "Thread-Splitter" logic to adapt long-form content for X (Twitter) character limits automatically.

Sentinel | Autonomous MLOps & Anomaly Detection Agent

- Designed and built a real-time monitoring microservice using FastAPI to proactively detect system anomalies and data drift in production machine learning environments.
- Implemented physics-based statistical thresholds (Z-Score via Scikit-Learn) to mathematically flag outliers in live time-series data streams with high accuracy and robustness.
- Integrated Google Gemini 2.5 as an autonomous "First Responder" agent to perform Root Cause Analysis (RCA) on error logs, demonstrating a novel approach to reducing mean-time-to-resolution.

- Deployed the full microservice architecture to the cloud (Render), establishing a publicly accessible REST API with Swagger UI documentation for seamless integration and developer access.
- **Tech Stack:** Python, FastAPI, Scikit-Learn, Google Gemini API, Render, Uvicorn

PolyMind: AI-Gated Quantitative Execution Engine

- Developed an institutional-grade, AI-gated arbitrage strategy for prediction markets, designed to overcome the limitations of latency-based bots by prioritizing execution quality.
- Engineered a sophisticated predictive model using XGBoost to forecast trade fill probability, actively mitigating liquidity traps and slippage in highly volatile order books.
- Implemented a robust quantitative execution engine capable of identifying and capitalizing on fleeting arbitrage opportunities in real-time, enhancing trading reliability and profitability.
- Leveraged Jupyter Notebooks for rapid prototyping, comprehensive backtesting, and rigorous validation of complex algorithmic trading strategies.
- **Tech Stack:** Python, Jupyter Notebook, XGBoost, Pandas, NumPy

TalentAlign | SaaS Recruitment Agent with Persistent Memory [Live SaaS]

- Developed a bias-free resume screening SaaS using Streamlit and Gemini 2.5 Flash, handling PDF parsing and skill gap analysis.
- Integrated Supabase (PostgreSQL) to store candidate metrics, enabling long-term tracking of hiring quality.
- Containerized the application with Docker to resolve deployment environment conflicts on Hugging Face Spaces.

NextClick | Real-Time Session Recommendation Engine [Big Data]

- Built a Self-Attentive Sequential Recommendation (SASRec) model using PyTorch to predict user intent based on session history.
- Processed the MovieLens 1M dataset to train a Transformer architecture capable of ranking 3,000+ items in <50ms.

Education

BSc Physics | University of Calabar, Nigeria

- Expected July 2025
- Focus: Quantum Mechanics, Computational Physics, Atmospheric Physics.
- Applied the Scientific Method to software debugging and experimental model validation.